Lucas Bucht

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Education

University of Pittsburgh — Pittsburgh, PA
Bachelor of Science in Electrical Engineering, Concentration in Integrated Circuits — *Expected May 2028*

Technical Skills

Analog & Small-Signal Design: Op-amp circuit design (LM386), signal path analysis, frequency response & filter/tone-shaping design, noise reduction & stability testing
Circuit Simulation & Layout: LTspice (SPICE simulation), Altium (schematic capture)
Lab & Debugging: Oscilloscope & multimeter-based debugging, protoboard prototyping & soldering
Digital Logic & HDL: VHDL, datapath design (ALU, registers, decoders, multiplexers), Siemens HDL Designer, ModelSim (testbench verification)
Embedded & FPGA: Quartus Prime, FPGA synthesis (DE10-Lite), Arduino IDE (C/C++)
CAD & Fabrication: Fusion 360, 3D printing, CNC, laser cutting
Tools: Git/GitHub, technical documentation

Projects

Guitar Fuzz Pedal | *LM386 Op-Amp, LTspice, Altium, Fusion 360* — Spring 2026
- Designed the full analog signal path using circuit analysis for an LM386 op-amp-based pedal
- Synthesized the amplification and frequency response in LTspice and Altium to verify component choices
- Prototyped and soldered on a protoboard, testing across 4 iterations to reduce noise and improve stability
- Created a scale enclosure in Fusion 360 to plan out jack placement and mounting points

VHDL Calculator | *Siemens HDL Designer, ModelSim, Quartus Prime* — Fall 2025
- Built a complete datapath in Siemens HDL Designer, including a 4x8 register and ALU
- Developed core components like an 8-bit register, 2:4 decoder, and multiplexer
- Wrote 10+ do-file testbenches covering all datapath components and verified functionality in ModelSim
- Synthesized the design to a DE10-Lite FPGA in Quartus Prime

Perovskite Solar Cell Research | *Technical Research & Writing* — Spring 2025
- Presented findings at the SSOE First Year Engineering Conference as lead writer of a 12-page analysis of the efficiency, cost, and sustainability advantages over silicon
- Utilized engineering research databases and technical writing standards to produce a professional academic report

UV-C Light Bar Waste Reduction System | *Fusion 360, Arduino IDE* — Fall 2024
- Designed a sustainable solution to reduce single-use disposable wipes in on-campus gyms
- Led CAD development across 5+ design iterations in Fusion 360, collaborating on a 3-person team

Experience

Engineering Makerspace Mentor — Sep. 2026 – Present
University of Pittsburgh — *Pittsburgh, PA*
- Trained students on safe operation of 3D printers, CNC machines, and laser cutters for project fabrication
- Enforced equipment safety protocols during 4+ hours of weekly open shop coverage
- Guided students through technical troubleshooting to keep projects on track

Panther Leadership Academy Peer Facilitator — Sep. 2026 – Present
University of Pittsburgh — *Pittsburgh, PA*
- Mentored 20 students on leadership philosophies, communication, conflict resolution, and group dynamics

Resident Assistant — Aug. 2025 – Present
University of Pittsburgh — *Pittsburgh, PA*
- Supported 40+ residents through conflict resolution, crisis response, and community engagement
- Organized 10 educational and social programs per semester to promote student success and well-being

Floor Staff — May 2025 – Aug. 2025
Regal Cinemas — *Crofton, MD*
- Provided high-volume customer service and supported theater operations during peak hours
- Mentored new hires by sharing operational knowledge and helping them acclimate to team workflows